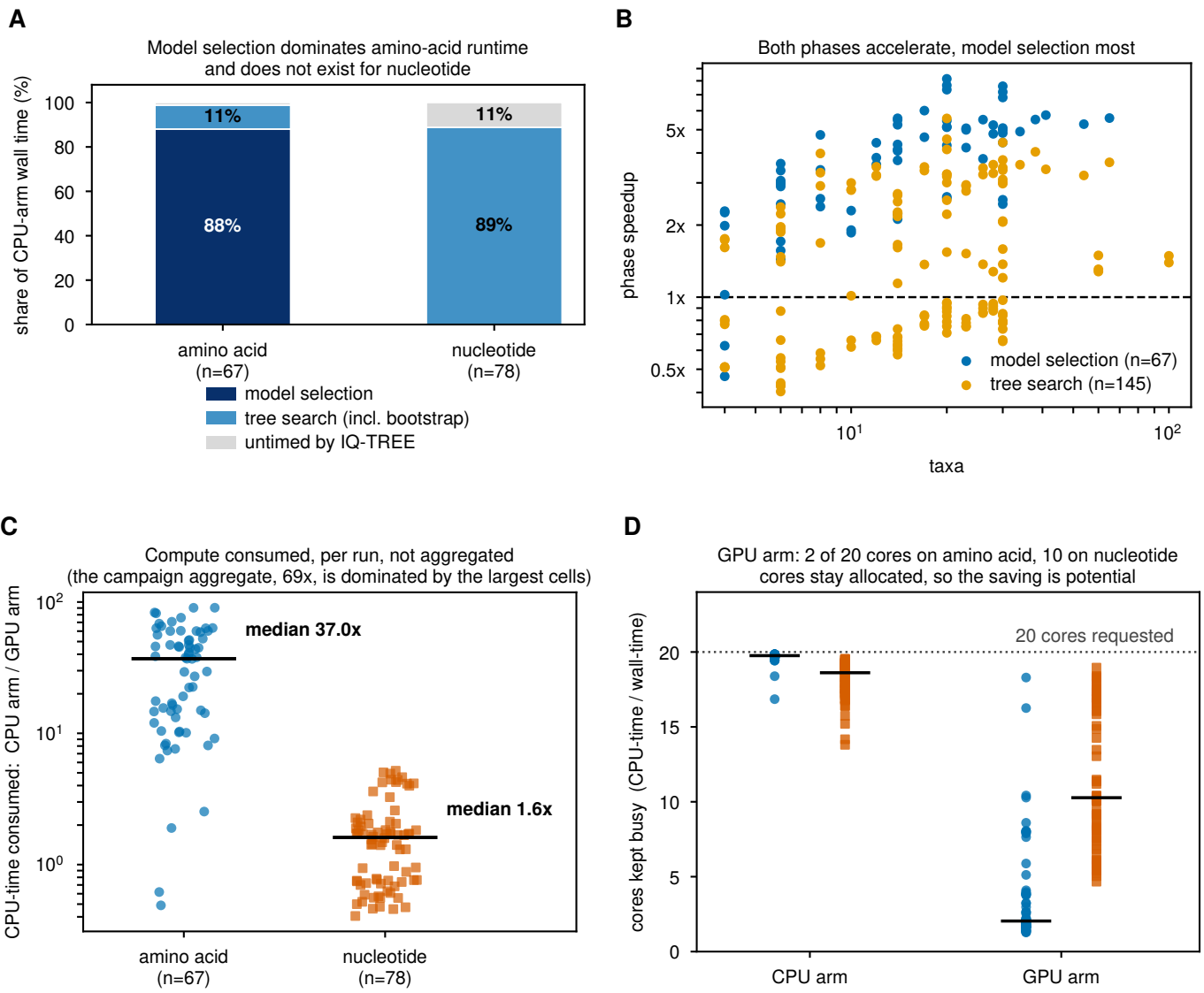

A
Model selection dominates amino-acid runtime
and does not exist for nucleotide
share of CPU-arm wall time (%)
100
80
60
40
20
0
11%
88%
11%
89%
amino acid
(n=67)
nucleotide
(n=78)
model selection
tree search (incl. bootstrap)
untimed by IQ-TREE
B
Both phases accelerate, model selection most
phase speedup
5x
2x
1x
0.5x
model selection (n=67)
tree search (n=145)
$10^1$
$10^2$
taxa
C
Compute consumed, per run, not aggregated
(the campaign aggregate, 69x, is dominated by the largest cells)
CPU-time consumed: CPU arm / GPU arm
$10^2$
$10^1$
$10^0$
median 37.0x
median 1.6x
amino acid
(n=67)
nucleotide
(n=78)
D
GPU arm: 2 of 20 cores on amino acid, 10 on nucleotide
cores stay allocated, so the saving is potential
cores kept busy (CPU-time / wall-time)
20 cores requested
20
15
10
5
0
CPU arm
GPU arm